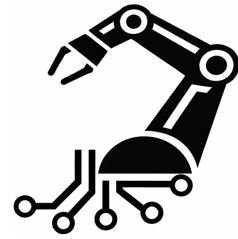


Suntracker Project

Final Documentation of a Theoretical Project



Author	Bartłomiej Dusza, Eng.
Date	June 17, 2026
Supervisor	Krzysztof Kwoka, Ph.D., Eng.

This document summarizes the project completed for the “Electronic Systems in Mechatronics” course.

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1 Project Objective and Assumptions

The objective of the project is to design a prototype energy storage system powered by photovoltaic panels. The device is intended to operate as a small autonomous energy source capable of charging a Li-Ion cell and powering a load through a 5 V section. Unlike simple fixed solar systems, the proposed device includes a mechanism that positions the panels relative to the direction of incoming light. This solution is intended to extend the effective exposure time of the panels and increase the amount of energy that can be harvested during the day.

Existing solutions of this type can be divided into several groups. The simplest photovoltaic systems use panels mounted at a fixed angle. Such a solution is inexpensive, simple and reliable, but it cannot actively adjust the panel position to the current position of the sun. More advanced designs use single-axis or dual-axis solar trackers that change the panel position during operation. Single-axis trackers most often follow the sun from east to west, while dual-axis trackers also allow the tilt angle to be corrected. This can increase the potential energy yield, but it also makes the mechanical design and control electronics more complex and worsens the energy balance because the actuators require power.

The proposed prototype uses an intermediate approach. Its purpose is not to maximize power at any cost, but to evaluate whether a small self-sufficient solar system is practical. The key assumption is a positive energy balance: the energy used for control, measurements and panel positioning should be lower than the additional energy obtained through improved panel positioning. For this reason, the system should operate periodically, remain in a low-power mode most of the time, and activate the actuators only when the panel position must be changed.

The design uses four small photovoltaic panels connected in series, a panel voltage and current measurement circuit, a DC/DC converter, a cell charging and protection circuit, and a microcontroller supervising the complete system. The system should measure the current power generated by the panels, adjust the operating point using a simplified pseudo-MPPT algorithm, and control the actuators responsible for positioning the structure. Safe charging of the Li-Ion cell and disconnection of actuators when they are not required are also important design requirements.

The scope of the project includes creating a prototype model, designing the power and control paths, preparing a mechanical structure that allows the panel position to be changed, and developing basic control software. The project is theoretical in nature.

2 Electronics

The system block diagram presents the general structure of the device's power, measurement and control paths. The photovoltaic panels are the energy source, and their voltage and current are measured to determine the current power supplied by the system. The energy is then transferred to the DC/DC converter and the BMS section responsible for charging and protecting the cell, which acts as the energy storage device. Separate voltage sections are supplied from the cell: 3.3 V for the logic and 5 V for the actuators, mainly the motors. The microcontroller is the central control unit. Based on the measured data, it supervises the pseudo-MPPT operation and controls the H-bridge responsible for positioning the panels.

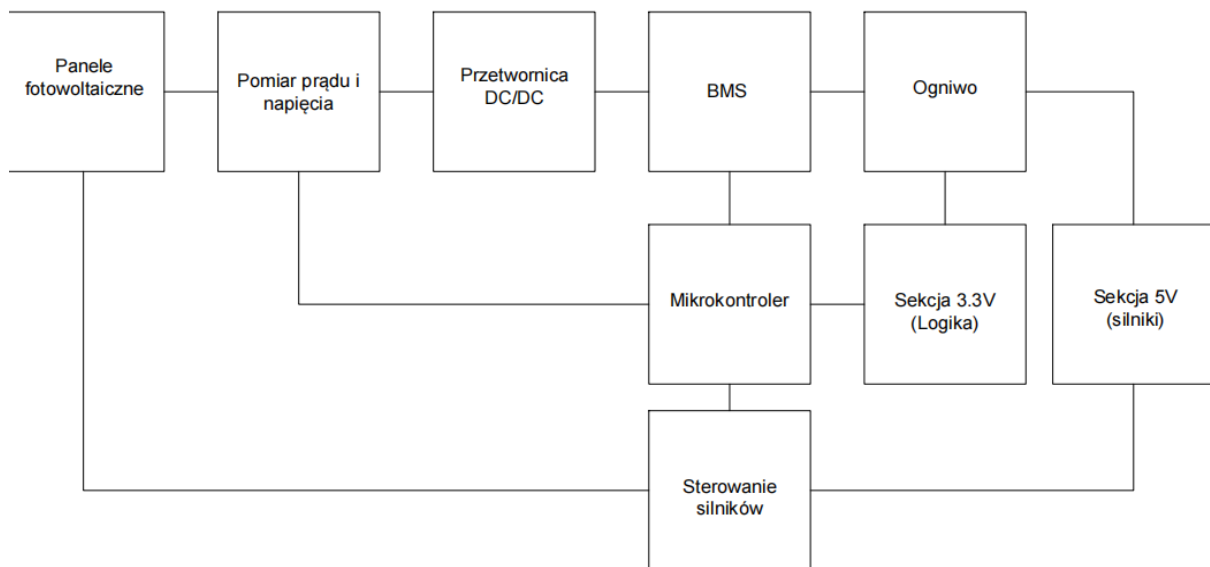


Figure 2.1: System block diagram.

2.1 Photovoltaic Panels

The power source is one of the most important parts of the system. Photovoltaic panels are nonlinear power sources that use the internal photoelectric effect. Because of this nonlinear behavior, maximum power point tracking (MPPT) is required to obtain the highest possible power. The main idea is to select the current drawn from the panel so that the system receives the maximum available power.

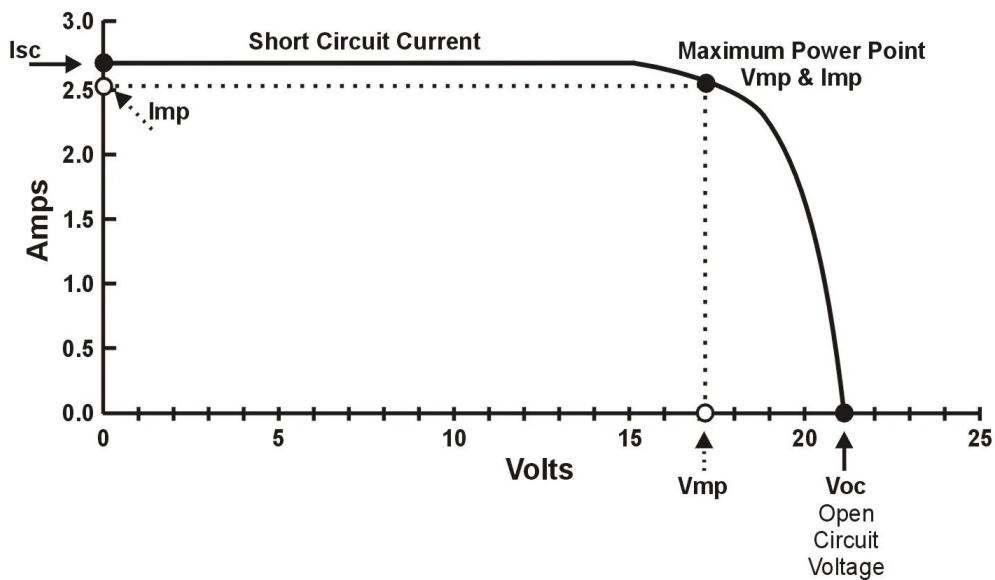


Figure 2.2: Panel current as a function of voltage [1].

Control of such a panel is based on reading its current voltage and current and then driving the DC/DC converter connected to the panel input. Photovoltaic panels can be connected in series to add their voltages, in parallel to add their currents, or in a mixed configuration. The project uses four identical panels, each with a nominal voltage of 5.5 V and a power rating of 0.4 W, connected in series. This configuration was selected to reduce transmission losses caused by the resistance of the wires. When connected in series, the panels provide a nominal voltage of approximately 22 V and a total power of 1.6 W. This provides enough voltage margin for the converter to operate under lower light levels. The disadvantage of this configuration is its sensitivity to differences in illumination between individual panels. If one panel is poorly illuminated, it may significantly limit the entire string. An alternative solution would be to connect two panels in series as one pair and then connect two such pairs in parallel, producing approximately 11 V at the output. The disadvantage is that current may flow from the better-illuminated pair to the less-illuminated pair. Schottky diodes can prevent this reverse current, but they introduce a voltage drop and additional power losses. With a series connection, bypass diodes may also be connected in parallel with the individual panels. They allow current to bypass a shaded panel or one that temporarily produces much less power than the others. As a result, one poorly illuminated panel does not block the complete string, although the total voltage is reduced. This improves resistance to partial shading, but increases the component count and introduces additional diode losses. All four panels could also be connected in parallel. In this case, the output voltage would remain approximately 5.5 V, while the nominal current would increase to approximately 0.29 A and the total power would remain approximately 1.6 W. This configuration would be more resistant to partial shading of one panel. However, such a low voltage would make it difficult to use a step-down converter and would most likely require a step-up converter. In practice, this option would be less suitable for the selected power architecture because it would produce higher wire currents and greater conduction losses.

2.2 Power Conversion

This section is divided into the converter used to implement pseudo-MPPT and the converters that generate the voltage rails required by the individual components from the battery voltage.

The converter section can be implemented in three main ways. The first is a custom discrete converter built from basic power-electronic components such as an inductor, diode, MOSFET and control circuit. The second approach is to use a ready-made DC/DC converter controller. The third option is a more highly integrated circuit that controls the converter and also provides additional functions such as MPPT and cell charging control.

For the proposed system, the main selection criteria were implementation complexity, ease of commissioning under prototype conditions, availability of documentation, and suitability for a photovoltaic-powered device. Table 2.1 compares the considered options.

Table 2.1: Comparison of converter section implementations

Solution	Advantages	Limitations	Assessment
Custom discrete converter	Very high design flexibility, component selection matched precisely to system requirements, and potentially high efficiency.	Highest design complexity, including the need to solve control, feedback, stability and power-component selection issues.	The most flexible solution, but too complex for the prototype stage
XL4015 [2]	Simpler implementation, sufficient voltage and current ratings, and basic built-in protection mechanisms.	Less flexible than a discrete solution and no native MPPT function.	The most practical choice for the prototype
BQ25750 [3, 4]	High level of integration, MPPT support and Li-Ion cell charging functions.	Greater application complexity, more difficult implementation and stricter requirements for the design of the complete power section.	The most functionally advanced solution

Considering the prototype nature of the device and the need to limit system complexity, the XL4015 was selected as the most reasonable solution [2]. Compared with the BQ25750, it is easier to implement and commission while providing sufficient parameters for the proposed system. However, the XL4015 does not provide native MPPT. This function must therefore be implemented externally, and the required cell charging profile must be handled separately.

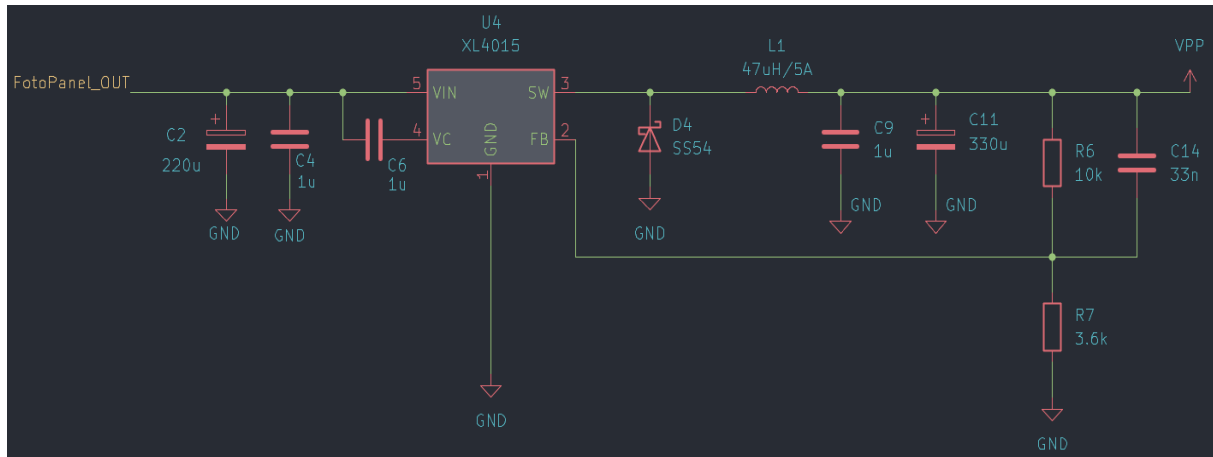


Figure 2.3: Converter circuit diagram.

The second section contains two converters: one supplies the 3.3 V rail and the other supplies the 5 V rail. The 3.3 V section uses the MCP1825S33. Its low dropout voltage is an important parameter because the regulator is powered directly from the battery. A power-path circuit is also used to switch the supply of this section automatically between the battery and the panels.

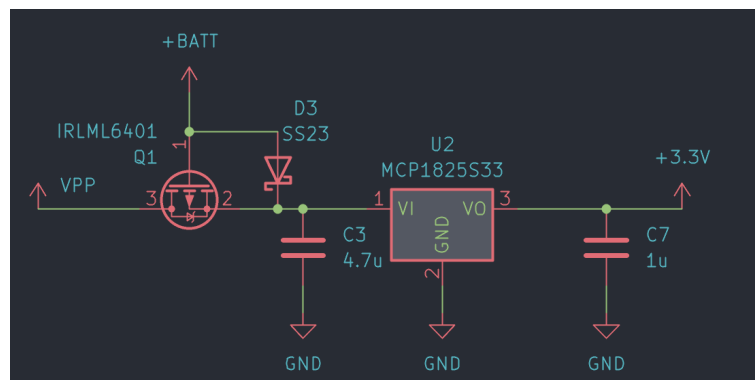


Figure 2.4: 3.3 V converter circuit diagram.

The second converter is based on the TPS61022, for which a low input voltage is also an important parameter. Another visible difference is the large number of capacitors. Their placement follows both the manufacturer's documentation and the needs of the system, because this section mainly powers the motors and supplies the device connected to the USB output. The capacitors are required because the DC motors and servos draw high currents during operation and introduce significant noise into these power lines.

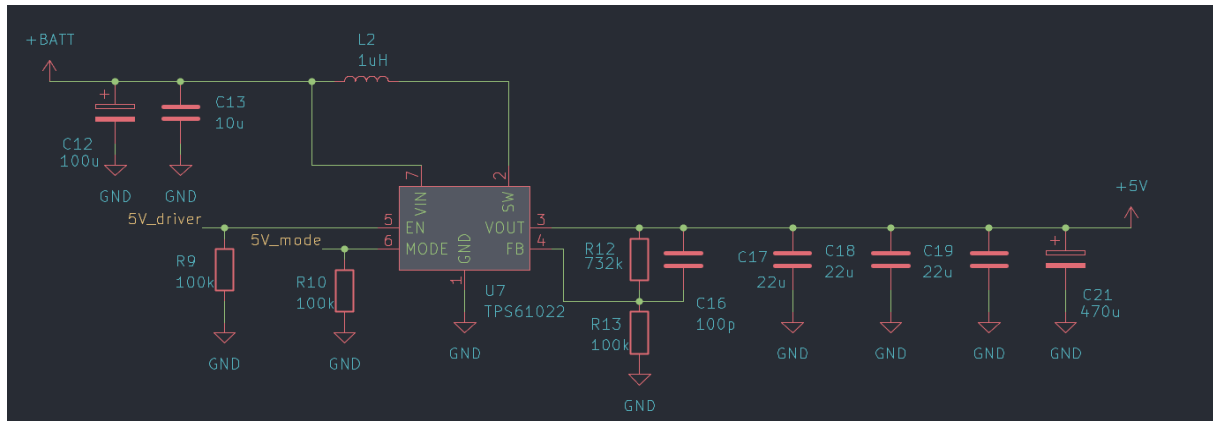


Figure 2.5: 5 V converter circuit diagram.

2.3 Microcontroller

Four representative solutions were analyzed during microcontroller selection: STM32L0, AVR128DB, ESP32-C6 and RP2040. The main requirements of the proposed system were low power consumption in sleep modes, reasonable consumption during active operation, peripherals required for energy measurement and actuator control, operation from a 3.3 V supply, and sufficient program memory.

Table 2.2: Comparison of selected microcontroller families for the proposed system

Family	Advantages	Limitations	Assessment
STM32L0 [5]	Very low power consumption in low-power modes, RTC and I2C/SPI/USART interfaces.	No native wireless connectivity and more complex configuration than simple 8-bit devices.	The best choice for the prototype
AVR128DB [6]	Very low standby power consumption, simple architecture and Sleep-Walking.	Lower performance, less memory and fewer functions than modern 32-bit devices.	A good option for a simplified version
ESP32-C6 [7]	Native Wi-Fi 6, Bluetooth LE, remote configuration and very low deep-sleep current.	High current during operation and transmission, which worsens the energy balance of the solar system.	Reasonable only when connectivity is required
RP2040 [8, 9]	High flexibility, two Cortex-M0+ cores, 264 kB SRAM and PIO.	Requires external QSPI memory and has unfavorable standby power consumption.	The least suitable choice for this project

The STM32L051C8 from the STM32L0 family was selected as the most suitable microcontroller for the prototype. It combines low power consumption with enough peripherals for the project, including an ADC, an I²C interface for communication with the INA219, timers, an RTC, and digital outputs for controlling power sections and actuators. Com-

pared with the STM32L4 family, the selected device offers lower performance and less memory. However, low power consumption, a simpler device structure, and sufficient functionality for the solar tracking and pseudo-MPPT algorithms were more important in this project. The AVR128DB family also has low power consumption, but it is less flexible than a 32-bit STM32 microcontroller [6]. The ESP32-C6 is an attractive solution when native wireless connectivity is required, but the high operating and peak currents of its radio make it less suitable from the energy-balance perspective [7]. The RP2040 is the least suitable device for a solar-battery application with long standby periods. The manufacturer specifies a current of approximately 180 μA even in deep sleep and recommends fully disconnecting the power supply in applications that require minimum standby current [8, 9].

The microcontroller section is based on the STM32L051C8Tx, which is the main control unit of the complete system. Programming and service connectors provide access to SWDIO, SWCLK, NRST and BOOT0. This allows the device to be programmed and debugged without removing the microcontroller from the PCB. NRST is pulled up to 3.3 V with a 10 k Ω resistor, while BOOT0 is pulled down to ground with a 10 k Ω resistor. As a result, the microcontroller starts the program from Flash memory after a reset. Power filtering is also an important part of this section. The supply lines use 100 nF decoupling capacitors and larger capacitors to reduce noise and temporary voltage drops during operation. The analog supply is additionally separated with a ferrite bead, which improves the resistance of ADC measurements to noise generated by the actuators and converters.

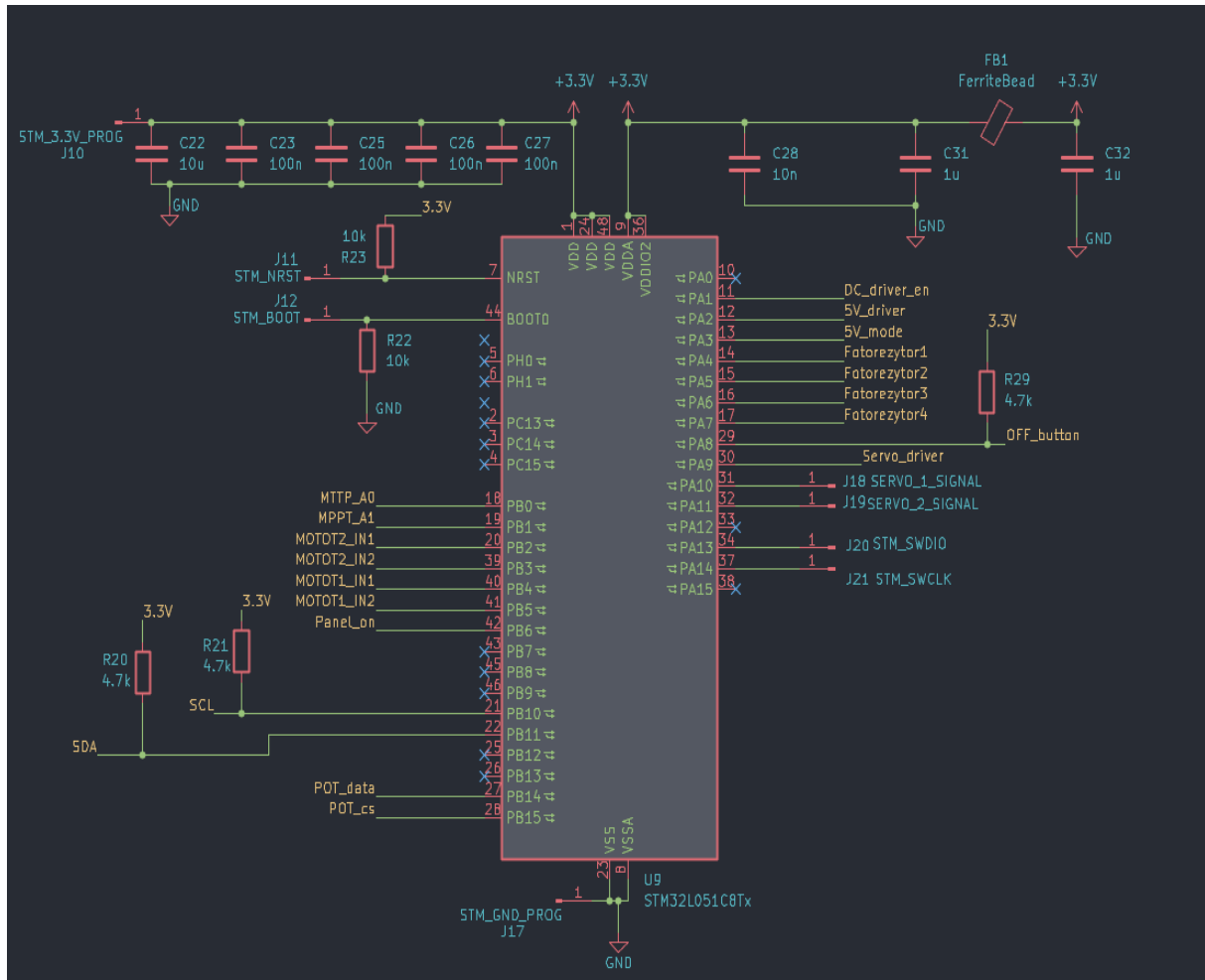


Figure 2.6: Microcontroller circuit diagram.

The microcontroller pins are divided by function according to the tasks performed by the system. PA0-PA3 control the power sections, including the motor driver enable, the 5 V supply and the converter operating mode. PA4-PA7 are analog inputs used to read the four photoresistors, which determine the direction of panel position correction. PB0 and PB1 are used for MPPT-related measurements, while PB2-PB5 control the direction of the DC motors. PB6 enables the panel section, and PB10 and PB11 form the I²C bus used, among other functions, to communicate with the INA219. PB14 and PB15 control the digital potentiometer used to adjust the pseudo-MPPT operating point.

2.4 Modules

2.4.1 Battery Management System (BMS)

The battery charging and protection section is based on the TP4056 and the DW01A protection circuit working with a dual MOSFET. The TP4056 charges a single Li-Ion cell, while the programming resistor sets the maximum charging current. LEDs indicate the charger status, including charging in progress and charge completion. The DW01A and MOSFETs protect the cell against over-discharge, overcharge and excessive current, allowing the battery to be used safely as the system's energy storage device.

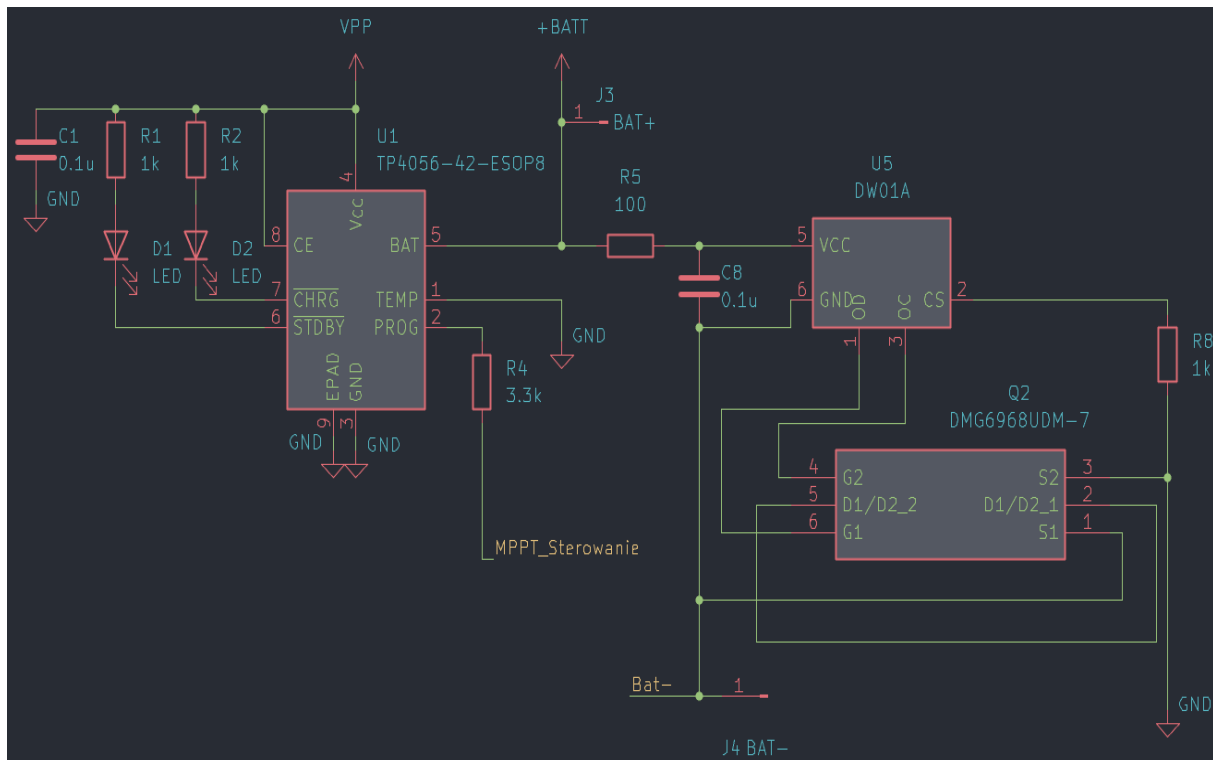


Figure 2.7: Cell charging and protection circuit diagram.

2.4.2 Operating Point Control

The design uses an MCP4021-5k digital potentiometer to adjust the operating point of the pseudo-MPPT circuit. The microcontroller controls the potentiometer through the POT_data and POT_cs lines and changes its resistance according to the currently measured panel power. The potentiometer wiper is connected to the MPPT_Sterowanie line, which affects the operating conditions of the power conversion section. This solution allows the load seen by the panels to be changed in software without a more complex MPPT controller.

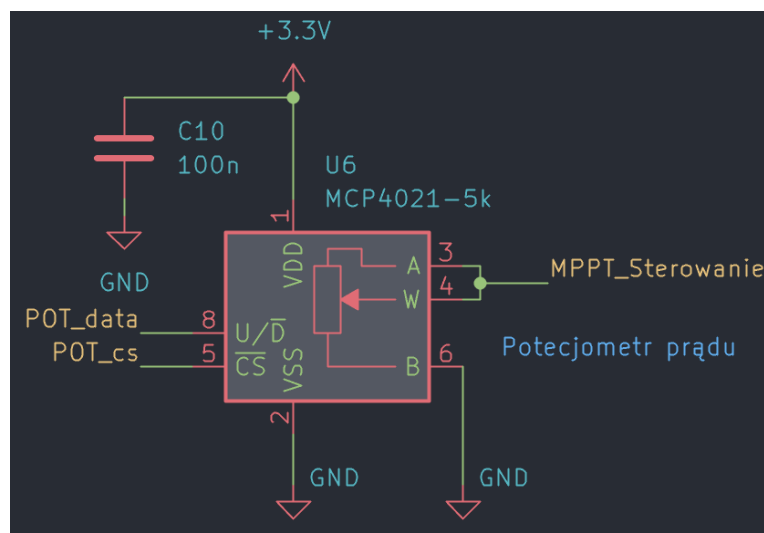


Figure 2.8: Digital potentiometer circuit diagram.

2.4.3 Panel Voltage and Current Measurement

The INA219AxD is used to measure the electrical parameters of the photovoltaic panels. The sensor measures the voltage drop across a shunt resistor in the panel power path, which allows the current drawn from the panels to be determined. It also measures the bus voltage, allowing the microcontroller to calculate the current power generated by the panels. Communication with the microcontroller uses the I²C bus, and a capacitor placed at the supply pin filters the power supply.

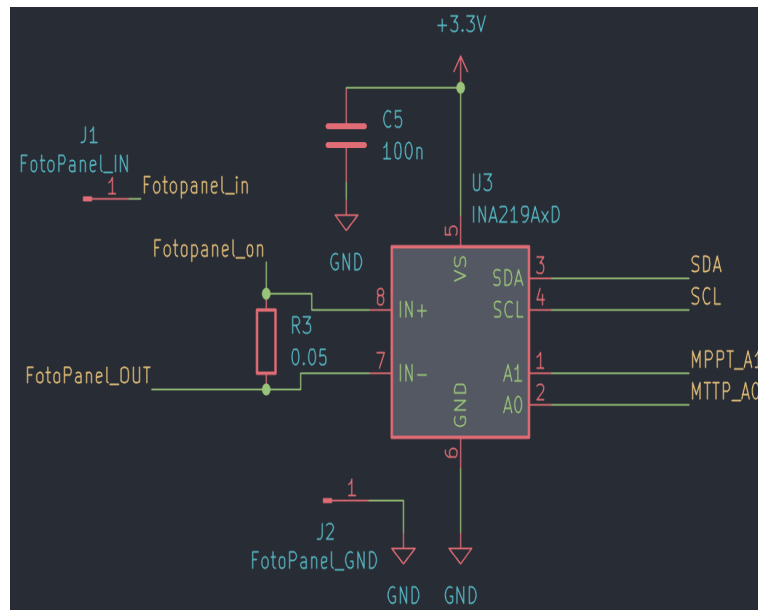


Figure 2.9: Power sensor circuit diagram.

2.4.4 Motor Control

The DC motor control section is based on the DRV8833PW dual H-bridge. It provides independent control of two motors through the MOTOT1_IN1, MOTOT1_IN2, MOTOT2_IN1 and MOTOT2_IN2 signals generated by the microcontroller. The DC_driver_en line enables the driver or puts it into sleep mode, reducing current consumption when the actuators are not in use. Low-value shunt resistors in the motor paths are used to limit and observe actuator current, while supply capacitors reduce the effect of noise generated by the motors. Because of the power dissipated in the shunt resistors, components rated for at least 0.5 W should be used in these positions.

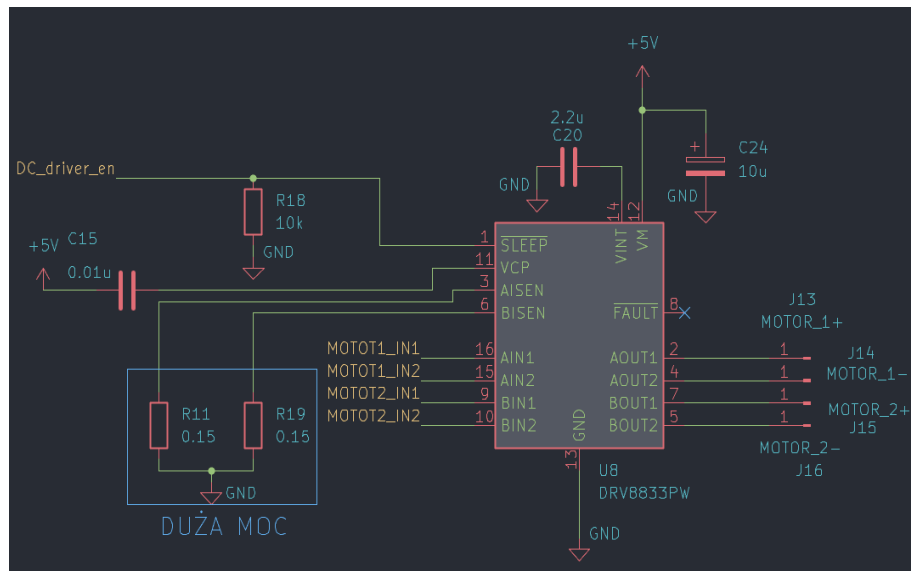


Figure 2.10: H-bridge circuit diagram.

R11 and R19 operate as current-sense shunts in the DC motor paths. Despite their low resistance, noticeable power may be dissipated in them during motor startup. They therefore require a higher power rating than typical signal resistors. The relationship $P = I^2 R$ means that even a short current increase significantly raises the thermal load on these components. The design also disconnects the servo supply with a simple high-side switch, so the servos do not draw current in sleep mode or when no movement is required.

3 Mechanical Design

The mechanical part of the project is a prototype structure that positions the photovoltaic panels relative to the light source and houses the main electronic components. The structure can be divided into two main parts: a base containing the electronics and drive section, and a moving fan-shaped panel assembly. This combines the enclosure, positioning mechanism and photovoltaic panel support in one compact structure.

The mechanical design is a prototype, so the main goals were to verify the concept, component placement and possible panel movement. The CAD model includes space for the PCB, motors, servo, transmission components, wiring and mounting points. Additional visualizations were prepared to assess component placement and potential collisions between moving parts.

3.1 Base

The base is designed as a cylindrical enclosure that supports the complete structure, protects the electronics and provides space for the drive components. It contains the PCB, power components, wires and part of the mechanism that rotates the panel assembly. The round shape suits the device because the structure can rotate relative to its surroundings and has no single fixed orientation.

The upper part of the base contains the PCB with the control and power circuits. The board is mounted on spacers to keep it away from the enclosure surface and reduce the risk of a short circuit. The PCB placement leaves space for wiring to the photovoltaic panels, motors, servos and external components such as the power button.



Figure 3.1: Interior of the base showing the PCB placement.

The enclosure also includes access and mounting holes. They allow user-accessible components to be placed outside, components to be installed, and the individual parts of the base to be joined. Clearance between the electronics and mechanical components is important so that movement of the transmission and shaft does not interfere with the PCB or wires. The final physical assembly is intended to use suitable heat-set threaded inserts.

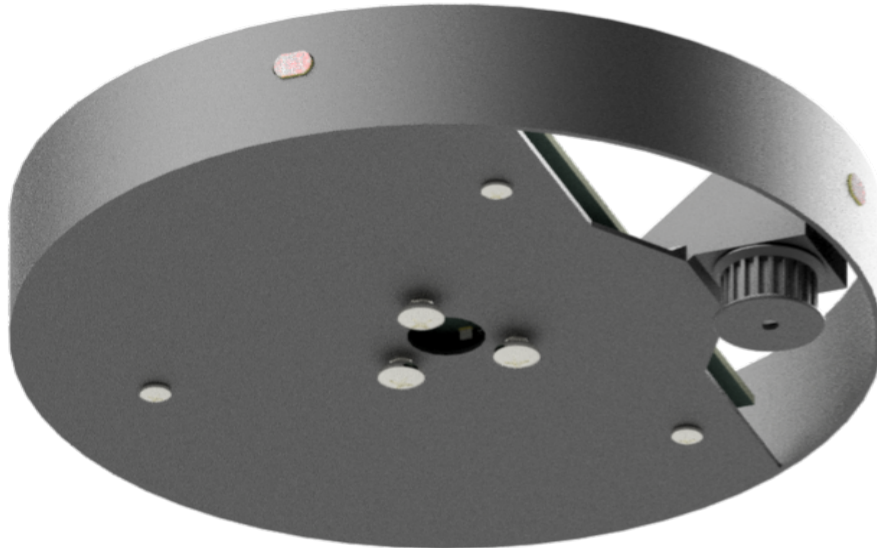


Figure 3.2: Lower side of the base and the space intended for mechanical components.

The base contains a mechanism that transfers motion to the main shaft. A transmission reduces the required motor torque and provides slower, more controlled movement. This is useful because a solar tracker does not require high rotational speed, but stable and repeatable panel positioning relative to the light source.

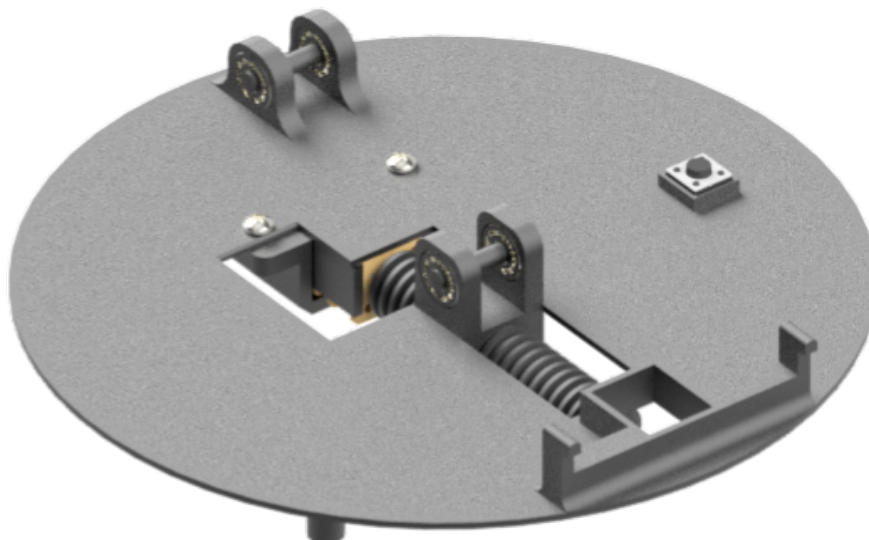


Figure 3.3: Upper part of the base with the drive mechanism components.

The drive section is placed below the upper plate of the base. This separates the electronics from moving mechanical components and makes better use of the available

space inside the enclosure. The model includes space for the motor, a belt transmission using a standard GT2 belt, and guiding components that transfer motion to the main axis of rotation.

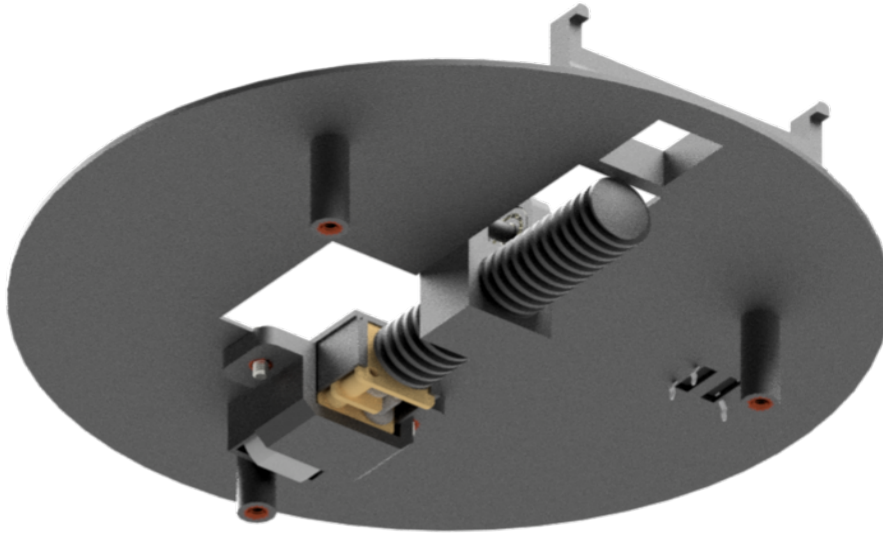


Figure 3.4: Drive mechanism viewed from the lower side of the base.

The lower part of the base closes the enclosure and provides space for the battery cells. The cells should be installed before the mechanism is assembled completely. The assembly order is important because access to the inside of the base becomes much more difficult after the gear has been installed on the main shaft.

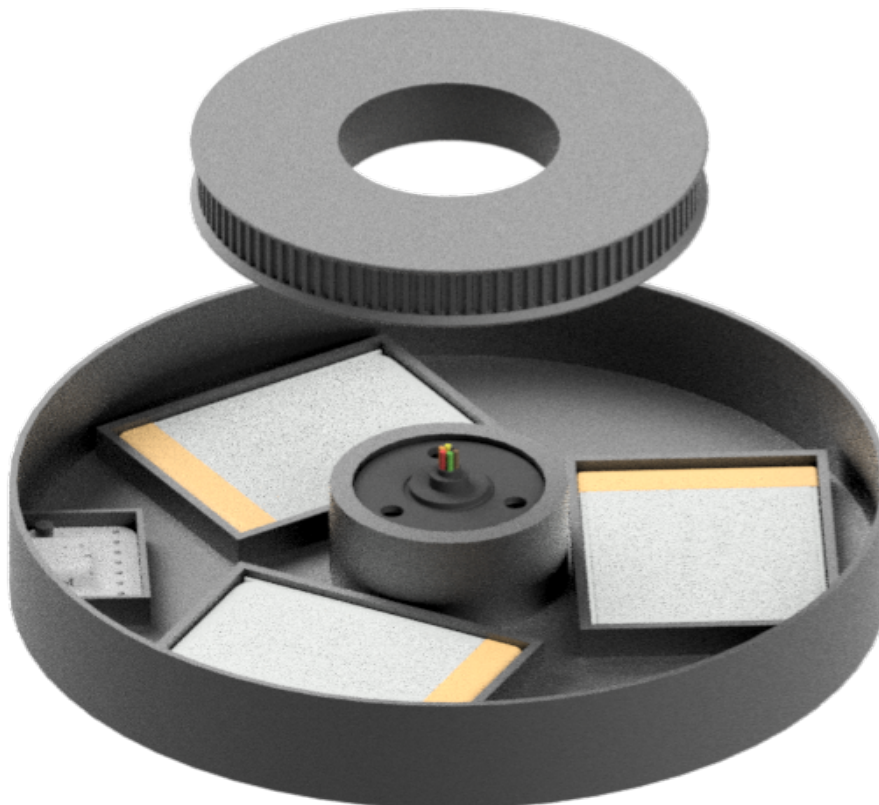


Figure 3.5: Lower part of the base with space for the cells and the main-shaft gear.

The gear on the main shaft transfers torque from the drive mechanism to the rotating upper part of the structure. In the planned assembly sequence, the cells and power wires are installed first, followed by the gear on the main shaft. In the final design, this component may use a tight fit on the shaft or be secured with adhesive to reduce the risk of slipping during operation.

The selected base design combines an electronics enclosure, cell mounting space and a support for the rotation mechanism. The main limitation of this version is difficult service access after complete assembly, which makes the assembly sequence important. A future version could use an easier-to-remove gear mounting method or a separate service cover that provides access to the cells without disassembling the mechanism.

3.2 Fan Assembly

The moving part of the device is a fan-shaped assembly consisting of four photovoltaic panels. The panels are arranged around a central mounting point, which increases the active surface area while keeping the structure compact. Compared with one large panel, this solution is more convenient for a prototype because it makes the panels easier to arrange and integrate with the folding mechanism.

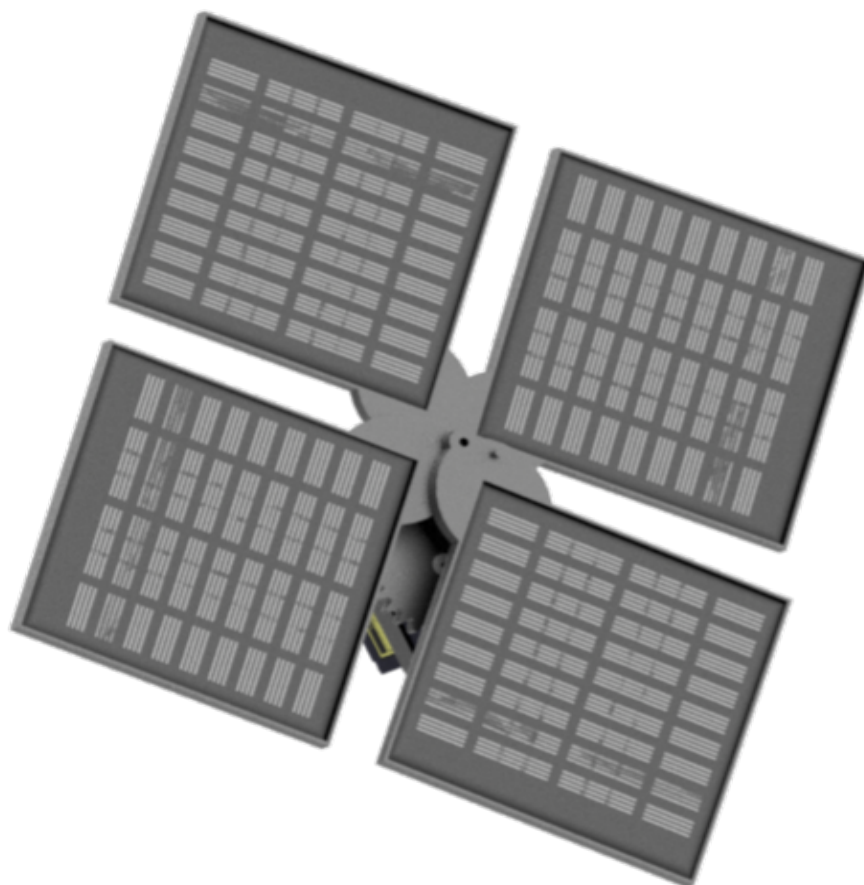


Figure 3.6: Moving assembly with four photovoltaic panels.

The panels are attached to a central mechanism that changes their position relative to the base. Its purpose is to place the panels at a more favorable angle to the light source.

The design uses a servo and a transmission that increases the useful range of motion. This allows the panel position to be controlled with a simple actuator.

The central panel mounting keeps the load relatively symmetrical. This is important because uneven mass distribution could increase the required actuator torque and load on the transmission. The model also provides space to route wires from the panels to the electronics in the base.

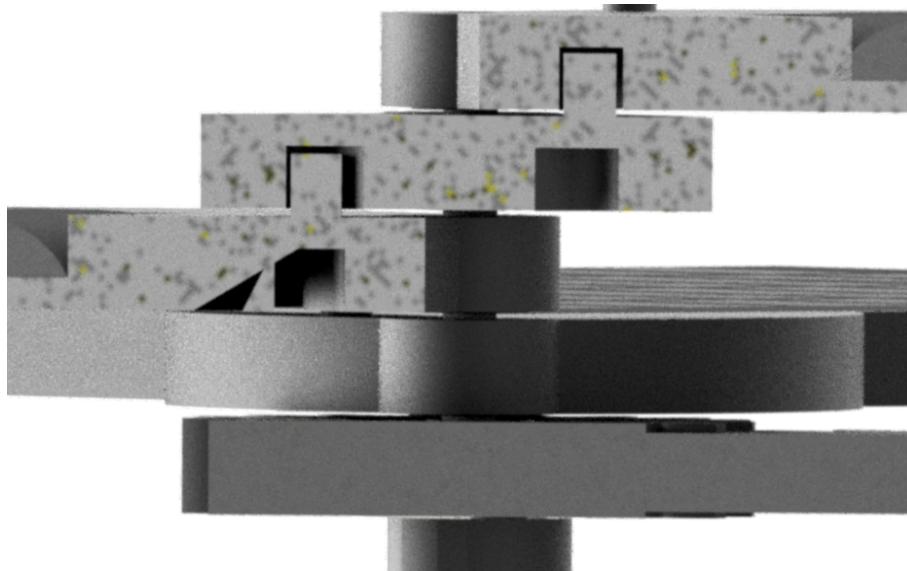


Figure 3.7: Close-up of the panel positioning mechanism.

Potential collisions between moving components were also checked during the design process. Particular attention was paid to the central joint, the transmission and the points where the panels pass the edges of the base. This verification is important because even a small displacement in a folding mechanism may limit the range of motion or block the structure. The movement of each fan segment is based on a round pin in the lower segment. Through a form-fit connection, the pin moves inside a channel in the upper segment.

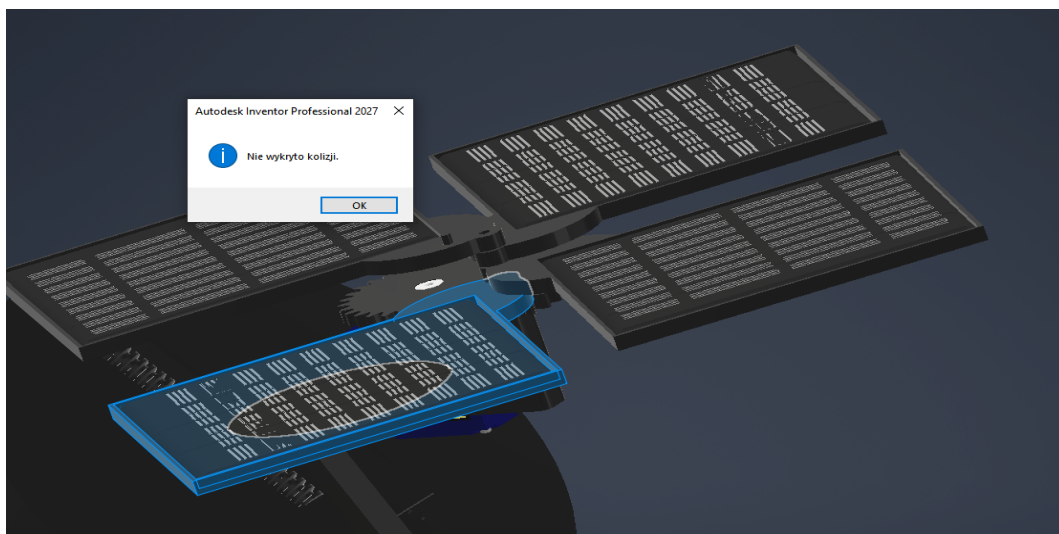


Figure 3.8: Analysis of the working space and potential collisions between moving components.

4 Software

4.1 Program Structure

The software is divided into modules corresponding to the main hardware blocks. The main file initializes the system and handles the state machine, while the remaining modules provide power measurement, actuator control, photoresistor and digital potentiometer support, and low-power modes. This structure simplifies software development and separates system control logic from functions that directly handle electronic components. The program was developed using the PlatformIO extension for Visual Studio Code. Because of the prototype nature of the project, the Arduino framework was used. For a final device, it would be more suitable to rewrite the code using a framework closer to the STM32 architecture.

The program operates cyclically, but it does not perform all measurements and movements continuously. Because the device is solar-powered, reducing the active time of the microcontroller and actuators was a priority. A state machine is therefore used to run individual functions only when they are required.

Table 4.1: Program modules

Module	Function
main.cpp	Main program loop and state machine handling.
General.cpp	Startup, sleep, folding and power-section control functions.
MPPT.cpp	Power measurement and panel operating-point control.
Photoresistors.cpp	Photoresistor readings.
Motors.cpp	DC motor and servo control.
Interrupt.cpp	Interrupt, power-button and periodic wake-up handling.

4.2 System State Machine

The main program logic is based on a state machine. This provides a clear definition of the current device mode and the actions that should be performed at a given time. The program contains five basic states: START, Harvest_update, Harvest, FOLD and Sleep.

The START state initializes the system after power-up or wake-up. It activates the required power sections, sets the initial mechanism position, performs initial panel positioning and initializes the pseudo-MPPT algorithm. When these operations are complete, the system enters the Harvest state.

Harvest is the main operating state. In this mode, the operating point of the photovoltaic panels is adjusted periodically based on voltage and current measurements. After the

update, the system enters sleep mode to reduce energy consumption.

The Harvest_update state repositions the panels according to the current direction of incoming light. It is entered periodically after a specified number of sleep cycles. In this state, the system reads the photoresistors again, corrects the panel position and updates the operating point.

The FOLD state folds the mechanism. It is entered when the system is switched off with the button. After the panels have been folded, the device enters sleep mode.

Sleep is the low-power state. Unnecessary power sections, including the motors and servos, are disconnected, and the microcontroller enters deep sleep. The system wakes after a specified time or when the button is pressed.

Table 4.2: Program operating states

State	Operation
START	System startup, power activation, panel positioning and MPPT initialization.
Harvest	Panel power measurement and pseudo-MPPT operating-point update.
Harvest_update	Periodic panel position correction toward the strongest illumination.
FOLD	Mechanism folding after the system is switched off.
Sleep	Low-power mode, actuator disconnection and microcontroller sleep.

4.3 Sun Position Tracking Algorithm

Four photoresistors are arranged so that illumination levels on different sides of the panel can be compared to determine the direction of the strongest light. The program takes several measurements from each photoresistor and calculates an average value, reducing the effect of noise and individual incorrect readings.

The direction in which the panel should rotate is determined from the differences between the readings. If the light level is too low or the differences are within the specified dead zone, the position is not corrected. This avoids unnecessary motor operation when a change in position would not provide a noticeable improvement.

After the initial position correction, additional fine adjustment is performed. The system makes small steps in one direction and compares the panel voltage before and after each movement. If the voltage increases, movement continues. If the result becomes worse, the mechanism reverses the last step. This allows a more favorable panel position to be found without complex astronomical calculations.

4.4 Pseudo-MPPT Algorithm

The project uses a simplified maximum power point tracking algorithm. The microcontroller reads the panel voltage and current with the INA219 and uses these values to calculate the current power generated by the panels. The operating point is adjusted by changing the resistance of the digital potentiometer, which affects the operating conditions of the power conversion section. The algorithm uses the perturb-and-observe method. After each resistance change, the current panel power is compared with the power measured in the previous step. If the power has increased, the adjustment continues in the same direction. If the power has decreased, the adjustment direction is reversed. Minimum and maximum resistance limits prevent operation outside the accepted range. If the panel voltage falls below a defined threshold, the algorithm increases the resistance and reduces the load on the panels. This prevents excessive loading of the photovoltaic source under low-light conditions.

4.5 Actuator Control

The panel positioning mechanism uses DC motors and servos. The DC motors are controlled by a dual H-bridge that allows the direction of rotation to be changed and the drive to be stopped. The program determines the motor operating time from the requested displacement or angle and then activates the appropriate control outputs.

To reduce sudden current changes, the motor does not operate continuously with one drive value. Movement is divided into acceleration, higher-speed operation and deceleration stages. This reduces mechanism jerking and limits overloads during panel positioning.

The servos position selected parts of the mechanism. Their power can be disconnected when they are not in use, reducing current consumption in standby mode and during operation when their position does not need to change.

4.6 Power Management and Sleep Mode

Because the photovoltaic panels provide limited power, energy management is an important part of the program. The system does not remain active continuously. After measurements, MPPT adjustment or position correction, it enters sleep mode. Before entering sleep, it switches off the power sections for the motors, servos and selected 5 V logic components.

The system can wake periodically after a specified time or when the power button is pressed. The button has two functions: it starts the system when it is off and begins the mechanism folding procedure when the system is active. This event is handled with an interrupt.

Sleep mode reduces the average current consumption of the device. This is especially important because the energy used by the microcontroller and actuators should not exceed the additional energy obtained through improved panel positioning.

5 Validation

5.1 Energy Balance Verification

Because the project is theoretical and prototype-oriented, energy validation was performed using approximate calculations. The purpose of this section is to verify whether the selected system architecture can operate as a self-sufficient energy source and charge the selected Li-Ion cell pack. The project uses four photovoltaic panels, each with a nominal voltage of 5.5 V and a power rating of 0.4 W. When connected in series, they provide a nominal voltage of approximately 22 V and a total power of:

$$P_{PV} = 4 \cdot 0,4 W = 1,6 W$$

This is the maximum power available under favorable lighting conditions. In practice, losses in the DC/DC converter, charging circuit, wires and protection components must be considered. Assuming an XL4015 converter efficiency of approximately 85% and an efficiency of approximately 75% for the remaining charging path, the power that can be delivered to the cell pack can be estimated as:

$$P_{CHG} = 1,6 W \cdot 0,85 \cdot 0,75 \approx 1,02 W$$

The project uses three 450 mAh Li-Ion cells connected in parallel. This gives a 1S3P pack with a total capacity of:

$$C_{BAT} = 3 \cdot 450 mAh = 1350 mAh = 1,35 Ah$$

With a nominal Li-Ion cell voltage of approximately 3.7 V, the energy stored in such a pack is:

$$E_{BAT} = 3,7 V \cdot 1,35 Ah \approx 5,0 Wh$$

With a charging power of approximately 1 W, the theoretical time required to recharge the pack can be estimated as:

$$t_{CHG} = \frac{5,0 Wh}{1,02 W} \approx 4,9 h$$

In practice, this time will be longer because panel power depends on the angle of incoming light, cloud cover, temperature, partial shading and the current operating point. In addition, the final stage of Li-Ion charging uses a decreasing current, so a full charge

may take longer than a simple energy-to-power calculation suggests. Nevertheless, the result shows that several hours of good sunlight per day should allow the system to recharge the selected cell pack.

For approximately 1.02 W delivered to the cells, the pack charging current can be estimated as:

$$I_{CHG} = \frac{1,02 W}{3,7 V} \approx 0,28 A$$

For a 1.35 Ah pack, this corresponds to a charging rate of:

$$\frac{0,28 A}{1,35 Ah} \approx 0,21 C$$

This is a low value, which is favorable for the Li-Ion pack. Cells connected in parallel should have similar parameters and a similar state of charge before they are connected to limit equalization currents between them.

The next stage of validation estimates the energy used by the actuators. The DC motors are controlled by the DRV8833. The motor paths contain 0.15 Ω shunt resistors R11 and R19, allowing the H-bridge to limit motor current. For the DRV8833, the current limit threshold is related to the voltage at the xISEN sense input. Assuming a threshold of approximately 0.2 V, the current limit can be estimated as:

$$I_{CHOP} = \frac{0,2 V}{0,15 \Omega} \approx 1,33 A$$

This means that during startup or a temporary mechanism stall, the motor should not draw its full short-circuit current without limitation. This improves system safety, although it also limits the maximum available actuator torque. At an operating current of 0.5 A, the power dissipated in the shunt resistor is:

$$P_R = I^2 R = (0,5 A)^2 \cdot 0,15 \Omega = 0,0375 W$$

At the 1.33 A current limit, the instantaneous power increases to:

$$P_{RCHOP} = (1,33 A)^2 \cdot 0,15 \Omega \approx 0,27 W$$

These values are acceptable for short periods of operation. However, the temperature of the shunt resistors and H-bridge should be verified in a physical prototype during actual mechanism movement.

The motor that rotates the complete mechanism is a geared motor with a nominal speed of approximately 100 RPM at 6 V. In this project, it operates from 5 V with an additional 6:1 reduction provided by a GT2 belt. Using simplified linear scaling with voltage, its no-load speed at 5 V can be estimated as:

$$n_{M1,5V} = 100 \text{ RPM} \cdot \frac{5}{6} \approx 83 \text{ RPM}$$

After the 6:1 reduction, this gives an output speed of approximately:

$$n_{OUT} = \frac{83 \text{ RPM}}{6} \approx 13,9 \text{ RPM}$$

In the software, movement of the rotation motor uses three PWM stages: acceleration, main movement and braking. For PWM values of 120, 240 and 20, the average duty factor is:

$$k_{PWM} = \frac{120 + 240 + 20}{3 \cdot 255} \approx 0,50$$

The simplified output speed with this drive pattern is therefore approximately:

$$n_{OUT,PWM} = 13,9 \text{ RPM} \cdot 0,50 \approx 6,9 \text{ RPM}$$

This corresponds to an angular speed of approximately 41 degrees per second. An example 90-degree position correction therefore takes approximately:

$$t_{M1} = \frac{90^\circ}{41^\circ/s} \approx 2,2 \text{ s}$$

At an operating current of 0.5 A, the energy used by one correction is:

$$E_{M1} = 5 \text{ V} \cdot 0,5 \text{ A} \cdot \frac{2,2 \text{ s}}{3600 \text{ s/h}} \approx 0,0015 \text{ Wh}$$

Even with 16 position corrections over several hours of operation, the energy used by the rotation motor is:

$$E_{M1day} = 16 \cdot 0,0015 \text{ Wh} \approx 0,024 \text{ Wh}$$

For an overload case in which the current reaches the approximate 1.33 A limit during movement, the energy used by 16 corrections would be approximately:

$$E_{M1CHOP} = 16 \cdot 5 \text{ V} \cdot 1,33 \text{ A} \cdot \frac{2,2 \text{ s}}{3600 \text{ s/h}} \approx 0,064 \text{ Wh}$$

This is still a small value compared with the energy stored in the cell pack, but such a condition should be treated as a short-term fault case and not as normal operation.

The second motor rotates the lead screw and moves the fan support linearly. The selected motor has a rated speed of approximately 410 RPM at 6 V, while the software

timing model uses 400 RPM. With a 5 V supply, the actual no-load speed may be closer to:

$$n_{M2,5V} = 410 \text{ RPM} \cdot \frac{5}{6} \approx 342 \text{ RPM}$$

For a screw lead of 10 mm/rev, this gives a linear speed of:

$$v_{LIN} = \frac{342}{60} \cdot 10 \text{ mm} \approx 57 \text{ mm/s}$$

An example 20 mm support displacement would therefore take approximately:

$$t_{M2} = \frac{20 \text{ mm}}{57 \text{ mm/s}} \approx 0,35 \text{ s}$$

At an operating current of 0.5 A, the energy used by such a movement is:

$$E_{M2} = 5 \text{ V} \cdot 0,5 \text{ A} \cdot \frac{0,35 \text{ s}}{3600 \text{ s/h}} \approx 0,00024 \text{ Wh}$$

Even with several support movements, this actuator has a negligible effect on the total energy balance. More important than the movement energy itself is whether the screw and guide mechanism produces excessive resistance. With a motor stall torque of approximately 1.3 kgcm at 6 V, scaled to 5 V, and an assumed screw and guide efficiency of approximately 30%, the short-term linear force can be estimated at approximately 20 N. A safer operating value with margin from stall is approximately 5 N. This means that the screw drive is sufficient for a lightweight mechanism, but it must be verified in a physical prototype, especially with regard to friction, guide misalignment and operation at the end positions.

The system also uses an MG90 servo. For a conservative case with a current of 0.75 A and a movement time of 1 s, the energy used by one servo movement is:

$$E_S = 5 \text{ V} \cdot 0,75 \text{ A} \cdot \frac{1 \text{ s}}{3600 \text{ s/h}} \approx 0,0010 \text{ Wh}$$

Even with ten such movements, the servo uses approximately 0.01 Wh. Limiting idle current is much more important than the energy used by one movement, so the design disconnects power from the servos and motor driver in sleep mode.

For an example operating cycle consisting of 16 rotation motor corrections, several support movements and several servo movements, the energy used by the actuators under normal operating conditions is approximately:

$$E_{DRIVE} \approx 0,036 \text{ Wh}$$

Compared with a pack containing approximately 5 Wh, this is less than 1% of its

stored energy. Under good lighting conditions, the panels could replace this energy within several minutes. Even in a short-term overload case in which the rotation motor periodically operates close to the H-bridge current limit, the energy used by the actuators remains much lower than the energy that can be harvested from the panels.

These calculations indicate that actuator energy consumption should not prevent self-sufficient system operation. Correct operation still requires limiting the number of movements, keeping the microcontroller in sleep mode between measurements, disconnecting actuator power after movement, and preventing prolonged motor or servo stalls. Actual current consumption during startup, steady movement and operation at the end positions should be verified in a physical prototype, because the real mechanical resistance will determine the actuator safety margin.

The calculations are approximate and do not include a detailed breakdown of every loss in the system. The energy balance includes the main losses in the conversion path through the assumed efficiency of the converter and remaining charging path. Detailed losses in wires, PCB traces, the H-bridge, switching components and other parts are not included. Energy used by the microcontroller during short active periods is also not calculated separately because its power consumption during actuator movement is much lower than that of the motors and servo. In sleep mode, the STM32L0 itself should draw so little current that it can be treated as negligible in a preliminary energy balance compared with the energy stored in the cell pack and used by the actuators. However, the sleep current of the complete PCB should be measured in a physical prototype. The actual standby current may be dominated not by the microcontroller, but by other system components such as converters, measurement circuits, pull-up resistors or indicators.

5.2 Mechanical Collision Verification

During analysis of the mechanical model, the placement of electronic components was checked relative to moving and drive components. The purpose was to determine whether the PCB and through-hole components would collide with the motors, transmissions or enclosure after assembly. The analysis identified a collision between one filtering capacitor and the motor located in the base.

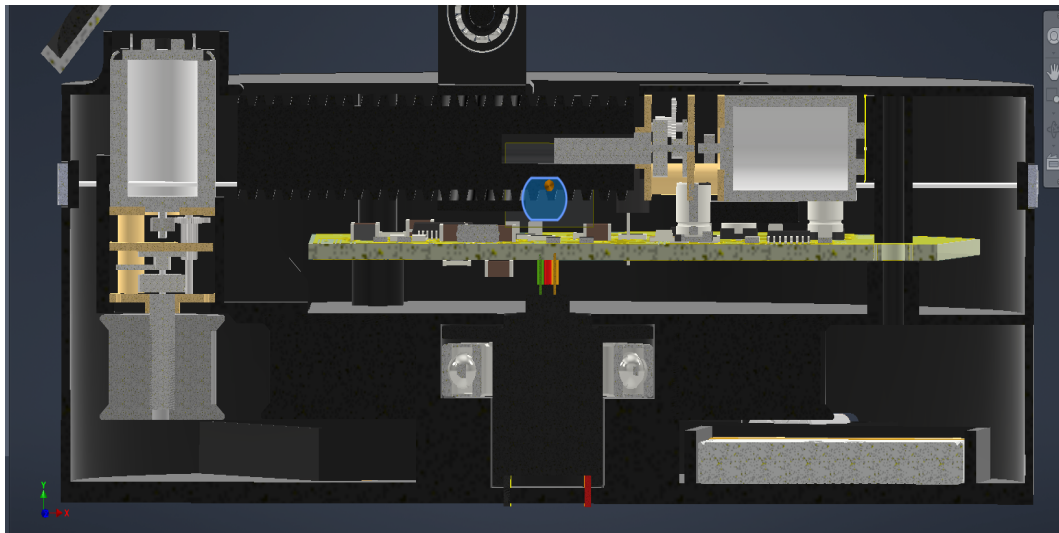


Figure 5.1: Close-up of the collision between the filtering capacitor and the motor.

The component involved provides filtering in the power path. Because the design also contains other decoupling and filtering capacitors, the simplest solution for this prototype version is to leave this particular capacitor unpopulated. Its footprint can remain on the PCB, but the component should not be soldered during assembly. This avoids the mechanical conflict without redesigning the complete board.

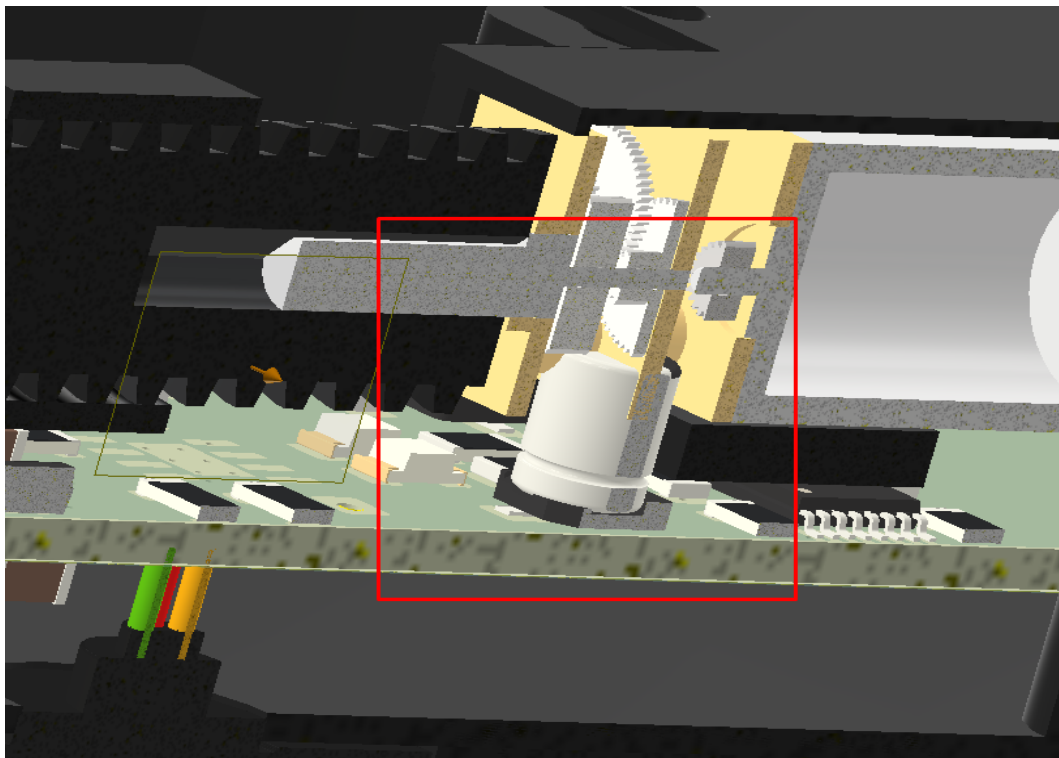


Figure 5.2: Device model cross-section showing the PCB, motor and mechanical components.

This component should be marked in the assembly documentation as DNP (do not populate). In a future PCB revision, the capacitor should be moved, replaced with a lower-profile component, or the motor placement should be changed. Leaving it unpopulated is acceptable in the current prototype because its function is supplementary

and the remaining filtering components still provide basic power-supply stabilization and decoupling.

5.3 Mechanical Verification of Actuator Selection

Mechanical validation estimates whether the selected DC motors are sufficient to move the prototype mechanism. The analysis is simplified because the actual actuator load depends on the quality of the mechanical parts, friction in the guides, transmission backlash, screw resistance and the exact distribution of panel mass. The purpose of the calculations is therefore to check the order of magnitude of the available torque and force, not to determine the exact dynamic behavior of the complete system.

The design uses two geared DC motors. The first rotates the complete mechanism around the vertical axis. It has a 298:1 gearbox, a rated speed of approximately 100 RPM at 6 V, a stall torque of approximately 4.0 kgcm and a stall current of approximately 1.6 A. The second motor rotates the lead screw and moves the fan support linearly. It has a rated speed of approximately 410 RPM at 6 V, a stall torque of approximately 1.3 kgcm and a stall current of approximately 1.6 A.

The motor that rotates the complete mechanism uses an additional 6:1 reduction provided by a GT2 belt. Because it is powered from the 5 V section, its available torque can be approximately scaled from the rated value at 6 V:

$$M_{1,5V} = 4,0 \text{ kgcm} \cdot \frac{5}{6} \approx 3,33 \text{ kgcm}$$

Converted to newton-metres:

$$M_{1,5V} \approx 3,33 \cdot 0,0981 \approx 0,33 \text{ Nm}$$

After including the additional 6:1 reduction and assuming a belt transmission efficiency of approximately 85%, the maximum short-term torque at the mechanism rotation axis can be estimated as:

$$M_{1,out} = 0,33 \text{ Nm} \cdot 6 \cdot 0,85 \approx 1,68 \text{ Nm}$$

This value corresponds to operation close to motor stall and should not be treated as continuously available torque. For a safer estimate, approximately 25% of this value can be used as the useful operating torque:

$$M_{1,rob} \approx 0,25 \cdot 1,68 \text{ Nm} \approx 0,42 \text{ Nm}$$

This torque should be sufficient to rotate the prototype mechanism because the actuator does not lift the complete structure but rotates it around the vertical axis. The main loads therefore result from bearing friction, transmission resistance and inertia during acceleration and stopping. The 6:1 reduction is useful because it increases the available

torque and reduces movement speed, improving control of the light-tracking system.

The output speed can be estimated in a similar way. For a 100 RPM motor at 6 V, scaled to a 5 V supply and including the 6:1 reduction, the result is:

$$n_{1,out} = \frac{100 \text{ RPM} \cdot \frac{5}{6}}{6} \approx 13,9 \text{ RPM}$$

The software also controls the motor with PWM, so the actual movement speed will be lower. This is suitable for the mechanism because a solar tracker does not require fast rotation, only periodic panel position corrections. Even if one correction takes several seconds, device functionality is not affected.

The second motor rotates the lead screw and moves the fan support. At 5 V, its stall torque can be estimated as:

$$M_{2,5V} = 1,3 \text{ kgcm} \cdot \frac{5}{6} \approx 1,08 \text{ kgcm}$$

Converted to newton-metres:

$$M_{2,5V} \approx 1,08 \cdot 0,0981 \approx 0,106 \text{ Nm}$$

In this case, the screw converts torque into linear force. Assuming a screw lead of 10 mm/rev and an efficiency of approximately 30% for the screw and guide system, the linear force can be estimated from:

$$F = \frac{2\pi\eta M}{p}$$

where η is the mechanism efficiency, M is the torque, and p is the screw lead. Substituting the values gives:

$$F_{2,max} = \frac{2\pi \cdot 0,30 \cdot 0,106 \text{ Nm}}{0,01 \text{ m}} \approx 20 \text{ N}$$

This is a short-term force corresponding to operation close to motor stall. A safer operating value can be taken as approximately 25% of this value:

$$F_{2,rob} \approx 5 \text{ N}$$

This means that the screw drive should be sufficient to move the lightweight fan support, provided that the mechanism does not have excessive movement resistance. In practice, this actuator is more sensitive to manufacturing quality than the main rotation drive. Guide misalignment, excessive screw friction or components binding at the end positions could significantly increase the required torque. For this reason, smooth support movement and motor current during motion should be checked carefully in a physical prototype.

Additional protection is provided by the DRV8833 H-bridge with $0.15\ \Omega$ shunt resistors R11 and R19. For this device, current limiting is activated when the voltage at the xISEN sense input reaches approximately 0.2 V. The current limit can therefore be estimated as:

$$I_{CHOP} = \frac{0,2\ V}{0,15\ \Omega} \approx 1,33\ A$$

This value is close to the expected motor stall current at 5 V. If the mechanism stalls temporarily, the H-bridge should therefore limit the current and reduce the risk of damage to the power path. This does not mean that the motors can remain stalled for a long time. Current limiting improves electronic safety, but it also limits the maximum available torque and does not eliminate heating of the motors, H-bridge and shunt resistors.

The estimates indicate that the selected motors are sufficient for the prototype mechanism. The main rotation motor has the largest margin because of the additional 6:1 reduction and should move the mechanism without difficulty if the rotation axis does not have excessive resistance. The screw motor should also be sufficient to move the fan support, although its performance depends more strongly on friction and the manufacturing accuracy of the guide components. The next stage of work should therefore verify the current consumption of both actuators during actual movement, startup and operation at the end positions. This would confirm whether the mechanisms operate below the H-bridge current limit and whether the selected motors provide a sufficient safety margin.

6 Final Conclusions

The project developed the concept of a prototype energy storage system powered by photovoltaic panels. It combines electronics, mechanical design and software, allowing energy to be harvested from the panels and their position to be adjusted actively relative to the light source. The main project assumption was a positive energy balance: the energy used by the control system and actuators should remain lower than the additional energy that can be harvested through improved panel positioning.

The analysis shows that the selected architecture is reasonable for a small autonomous solar system. Four photovoltaic panels connected in series provide sufficient voltage margin for the DC/DC converter, and the 1S3P Li-Ion cell pack can operate as a small energy storage device. The energy-balance calculations indicate that the system should be able to recharge the selected cell pack under favorable lighting conditions. However, it should not be treated as a full-size, high-capacity powerbank. It is a prototype small energy source intended to power low-power loads or partially recharge devices with higher current consumption.

An important part of the project is the pseudo-MPPT algorithm and panel voltage and current measurement. This solution allows the current panel power to be evaluated and the operating point to be changed without a more complex dedicated MPPT controller. The approach is simpler and fits the prototype nature of the project, but its effectiveness must be tested under real lighting conditions. A comparison of system operation with and without active solar tracking would be particularly important.

The mechanical design meets the prototype assumptions because it houses the electronics and cells, supports the actuators, and allows the panel position to be changed. The fan structure increases the active panel area while keeping the device compact. At the same time, the mechanical design requires further verification after a physical model has been built. The stiffness of printed components, transmission backlash, movement resistance, panel positioning accuracy, and ease of cell assembly and servicing should be checked in practice.

The software has a modular design and is based on a state machine. As a result, device operating modes such as startup, light tracking, pseudo-MPPT adjustment, folding and sleep are separated logically. Periodic operation and entering sleep mode after measurements or position corrections are especially important for the energy balance. This reduces average current consumption and increases the possibility of self-sufficient system operation.

The main limitation of the current project version is the lack of a built and tested physical prototype. The validation is theoretical and is based on calculations, circuit analysis, the CAD model and assumed component parameters. The next development stage should therefore be construction of the prototype, commissioning of the power path, measurement of the actual charging current, verification of sleep-mode current consumption, and evaluation of the energy used by the actuators during actual panel

positioning.

Further work should also verify the actual converter efficiency, the temperature of power-path components, system behavior under partial panel shading, and pseudo-MPPT algorithm stability. Only these tests would allow a clear assessment of whether active panel positioning provides a noticeable energy gain compared with a fixed structure. Despite these limitations, the project provides a consistent basis for building a physical prototype and further verifying the operation of a small self-sufficient solar system.

7 References

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Appendix A - PCB Layout

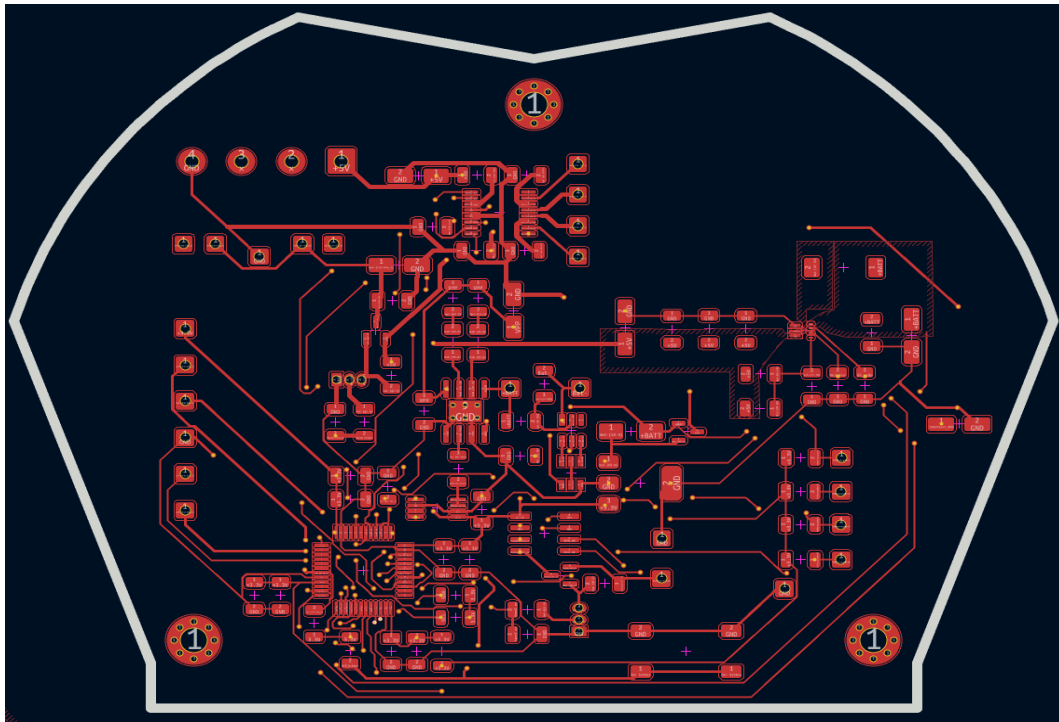


Figure 7.1: Front PCB layout.

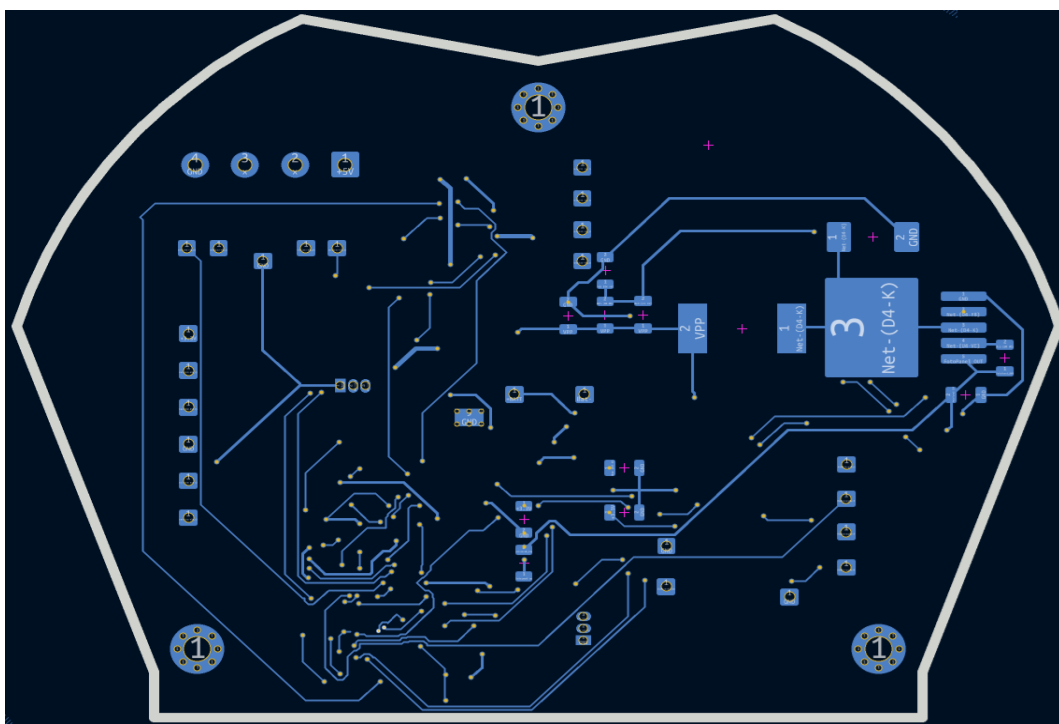


Figure 7.2: Back PCB layout.